

REINFORCEMENT LEARNING

The Training of a Digital Champion

Teaching AI through **Trial and Error**

Planrs Academy | Grade 6 Session 1

THE FLAPPY BIRD MYSTERY

Can a Robot Play Games?

Imagine watching a robot play a video game like **Flappy Bird**. At first, the robot is terrible. It hits the very first pipe and crashes. It fails 10 times, 50 times, even 100 times!

But then, on the 101st try, something amazing happens. The robot clears one pipe... then two... then a hundred! Suddenly, it's a pro player. How did it get better without a human telling it the rules?

The Answer: The robot used **Reinforcement Learning**. It learned by making mistakes and receiving "rewards" for doing well.

Today, we are going to learn how we can turn a "clumsy" AI into a **Digital Champion**.

WHAT IS REINFORCEMENT LEARNING?

Learning by Doing

Reinforcement Learning (RL) is a type of machine learning where an AI learns to make decisions by trying different things and seeing what happens.

It's just like how you learned to ride a bicycle:

- You tried to pedal (Action).
- You fell down (Penalty/Pain).
- You tried again with more balance (New Action).
- You stayed upright and moved forward (Reward/Fun!).

Over time, your brain remembered the "good" actions and stopped the "bad" ones. An AI does the exact same thing!

THE THREE PARTS OF RL

Meet the Agent, Action, and Environment

To teach an AI, we need to understand the three main characters in our story:

1. The Agent (Sheru the Puppy)

The **Agent** is the AI learner. Imagine a puppy named **Sheru**. Sheru is the one who is learning a new trick.

2. The Environment (The Park)

The **Environment** is the world the agent lives in. For Sheru, this is the park where he runs and plays.

3. The Action (Shake Hands)

The **Action** is what the agent chooses to do. Sheru might choose to bark, sit, or try to "Shake Hands" with you.

THE FEEDBACK LOOP

Shabash vs. Oops!

How does the Agent know if an Action was good or bad? We use **Rewards** and **Penalties**.

The Cricket Practice Example:

Imagine a young cricketer practicing his shots with a coach:

- **Good Action:** He hits the ball through a gap for a boundary.
→ **Reward:** The coach shouts "**Shabash!**" and he gets +4 runs.
- **Bad Action:** He hits the ball straight to a fielder.
→ **Penalty:** He gets 0 runs and has to try again.

The cricketer wants to hear "Shabash" as much as possible, so he learns to hit the ball into the gaps!

MAXIMIZING THE SCORE

The Total Reward

An AI doesn't just want one reward; it wants the **highest total score** possible. It thinks like a gamer trying to beat a high score.

The AI's Internal Logic:

"If I move Left, I get 0 points."

"If I move Right, I get a star worth 10 points!"

Decision: "I will always move Right to get the most points!"

In games like **Temple Run** or **Subway Surfers**, the AI learns that jumping over a hurdle is better than hitting it because jumping keeps the game going and keeps the score growing.

LEARNING FROM MISTAKES

The Mela Adventure

The AI doesn't know the rules when it starts. It uses **Trial and Error** to explore. Imagine you are visiting a big **Mela** (Fair) for the first time:

- **Trial 1:** You turn left at the giant wheel and find a dead end. (Penalty)
- **Trial 2:** You turn right and find the **Jalebi Stall!** (Big Reward: +10 Inaam)

The next time you go to the Mela, you won't turn left. You'll go straight for the Jalebis! The AI does this thousands of times per second until it finds the perfect path.

THE ROAD TO SUCCESS

Why Does AI Fail 100 Times?

You might wonder: "Why is the AI so slow to learn at the start?"

It's because the AI has to map out the entire environment. It needs to know exactly which "Oops" leads to a penalty and which "Shabash" leads to a win. Every failure is actually a **data point** that makes the AI smarter.

The 100 Failures = 100 Lessons learned.

Real-World Superpowers

Reinforcement Learning isn't just for Flappy Bird. It's used for some of the coolest technology today:

Use Case	How it Uses RL
Self-Driving Cars	Learning to stay in lanes and stop at red lights to get "Safety Rewards."
Robotic Arms	Learning how to pick up fragile eggs without breaking them through practice.
Personal Assistants	Learning which news stories you like based on what you click on.

SUMMARY

The RL Toolkit

- **The Agent:** The AI "Puppy" that is learning.
- **The Action:** The moves the Agent makes.
- **The Reward (Inaam):** The positive feedback for a good move.
- **The Penalty (Saza):** The negative feedback for a mistake.
- **Trial & Error:** Trying different paths to find the one with the most rewards.

Success Criteria

Can you identify the Agent and Reward in a game of Ludo? Can you explain why failing in a game is good for an AI? If yes, you are a Reinforcement Learning expert!

FINAL MISSION

The Digital Coach

Before you go, think about this:

"If you were teaching an AI to play Hide-and-Seek, what would be the 'Big Reward' and what would be the 'Penalty'?"

Write your answer down and share it with your teacher. Great job training your digital champion today!